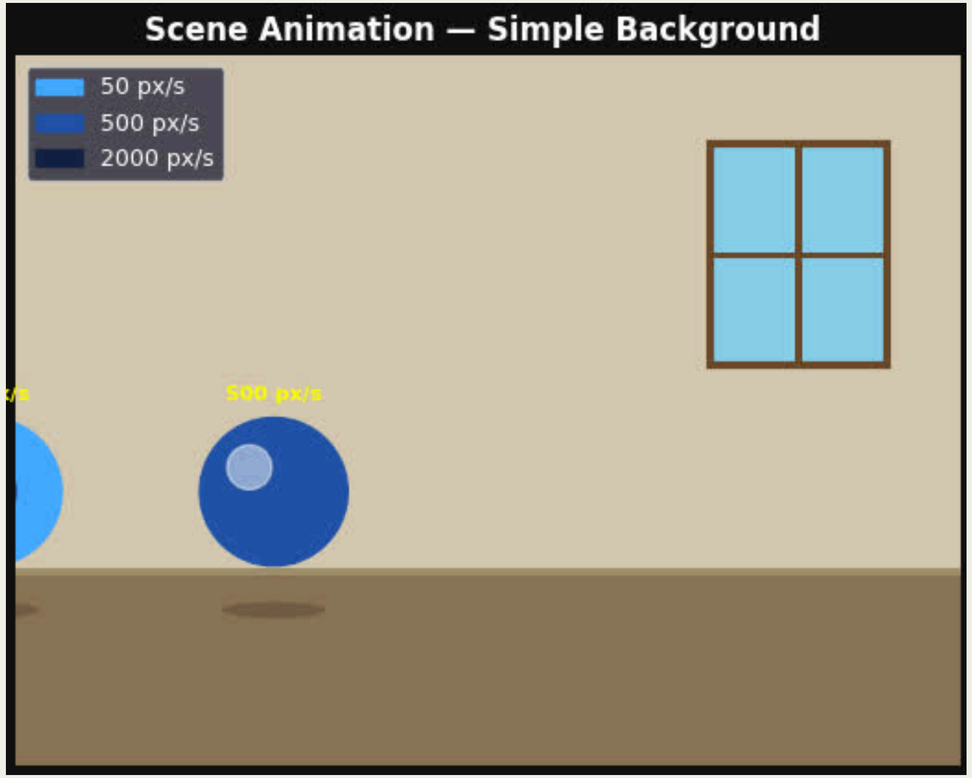
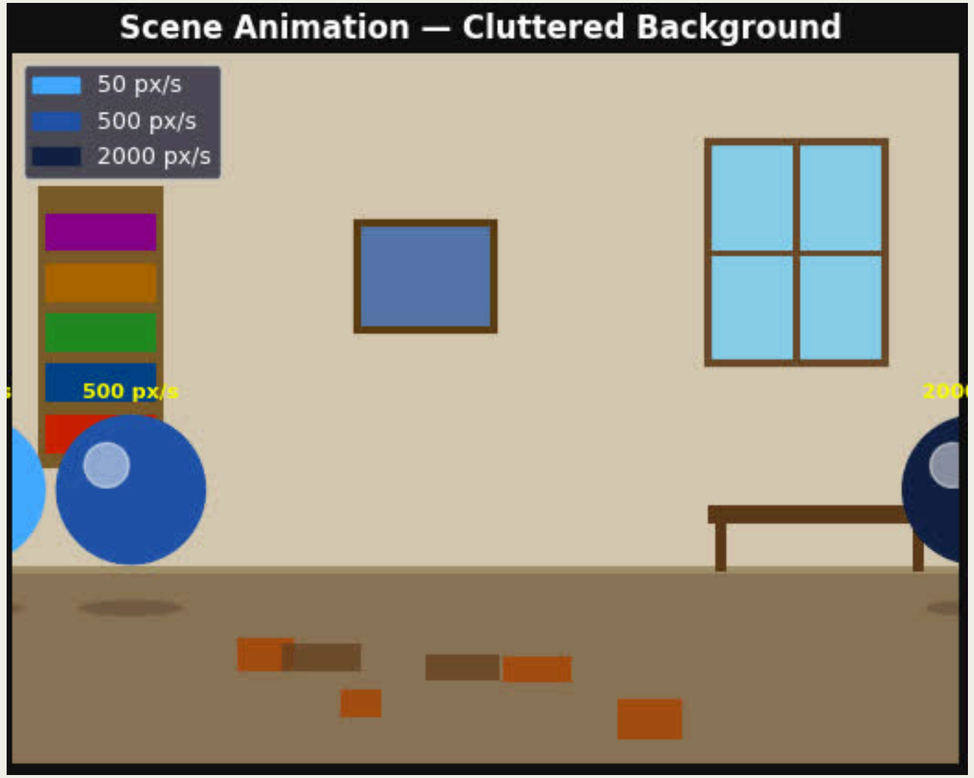
NOVEL IMAGE SENSOR MODELING EXTENSION

TEAM 4

SCENE MODEL





- EXTENDED VELOCITY RANGE FROM 500 PX/S TO 2000 PX/S FOR IMPROVED DVS POWER VISIBILITY
- REPLACED STATIC ROOM GRAPHIC WITH A SPHERE THAT MOVES AT THREE VELOCITIES (50, 500, 2000 PX/S) ACROSS BOTH SIMPLE AND CLUTTERED BACKGROUNDS
- ADDED TEMPORAL VARIATION—MODELS OBJECT CHANGING SPEED OVER TIME (NOT JUST CONSTANT VELOCITY), SHOWING DVS EVENT RATE RISING AND FALLING WHILE CIS REMAINS FLAT.

CIS MODEL: MODUCIS UPDATES

- **EXTENDED VELOCITY RANGE** TO 2000 PX/S (WAS 500 PX/S) TO MATCH UPDATED SCENE MODEL AND MAKE POWER VARIATION VISIBLE ACROSS SCENES
- **DUAL SCENE ANALYSIS —** FULL SCENE SWEEP [10→2000 PX/S, ALL OBJECT SIZES] AND SEPARATE ANIMATION SCENE ANALYSIS [50, 500, 2000 PX/S, OBJ=50PX] — EACH WITH THEIR OWN CSVS AND PLOTS
- **CAMERA VS CIS SNAPSHOT ANIMATION —** INTERACTIVE VISUALIZATION BUILT FROM SCENE MODEL ROOM, SHOWING SPHERE FREEZING BETWEEN CIS CAPTURES WITH LIVE EVENT RATE GRAPH PROVING CIS POWER STAYS FLAT AS SCENE ACTIVITY RISES
- **SCENE MODEL FULLY DRIVES CIS —** VELOCITIES, BACKGROUNDS, OBJECT SIZES AND ANIMATION PARAMETERS ALL READ AUTOMATICALLY. ANY UPDATE TO SCENE MODEL REFLECTS IN CIS ANALYSIS INSTANTLY

DVS MODEL: EVENT BASED SENSOR (UPDATES)

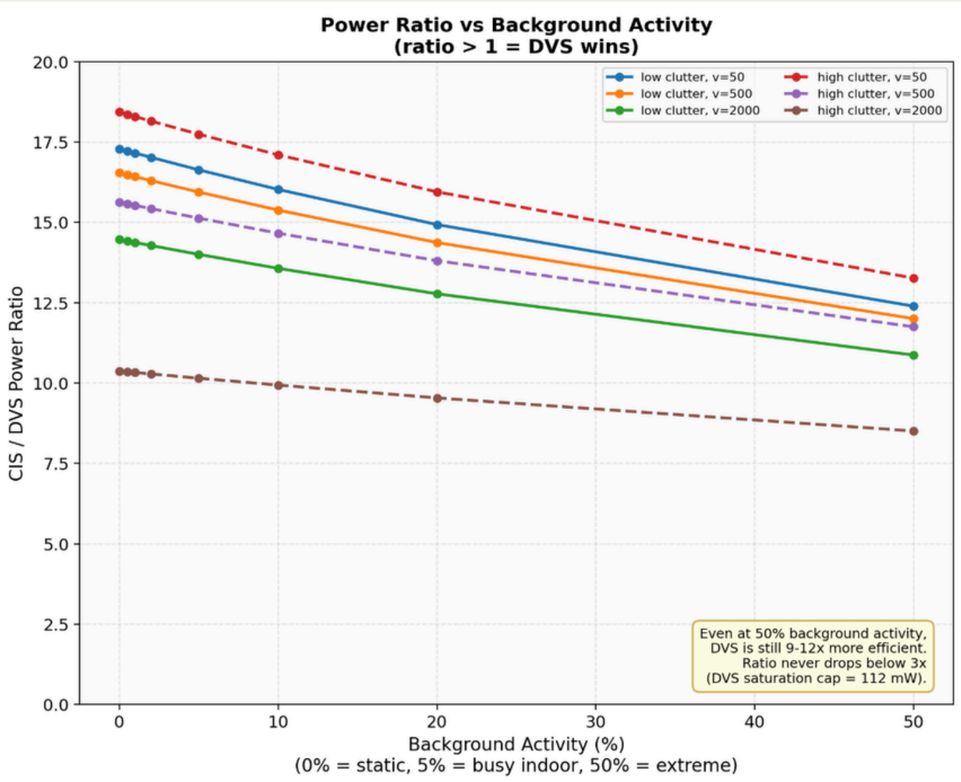
- **Threshold sweep:** sweeps θ across a range from 0.10 to 0.40 to show the sensitivity/noise tradeoff (lower θ = more events = more power but catches faint edges). Range of 0.1 to 0.5 given in Lichtsteiner 2008 paper. Every scene is now computed at three sensitivity levels instead of one callibrated value (0.20). Uses low_threshold = 0.1 for power comparison. - this is the % change in brightness intensity for an event to be recorded and for a pixel to be considered 'fired'.
- **Temporal variation:** added run_temporal_variation() which simulates tracking the object over time as the object starts slow, speeds up, peaks, then slows back down. V1 only computed power at fixed velocities.
- **Pixel firing breakdown:** added compute_pixel_breakdown() which computes active vs silent pixels, with background pixels scaling with velocity. Breakdown is displayed in a donut chart.
- **Velocity range extended:** at 500 px/s the dynamic power change was too small to see on a plot, hence the velocity range was extended to 2000px/s.
- **Object-tracking flag:** added dvs_can_track to the output CSV, this checks whether the event rate is sufficient to detect the object edge at least once per crossing time and confirms DVS can track in all scenes in our parameter space.

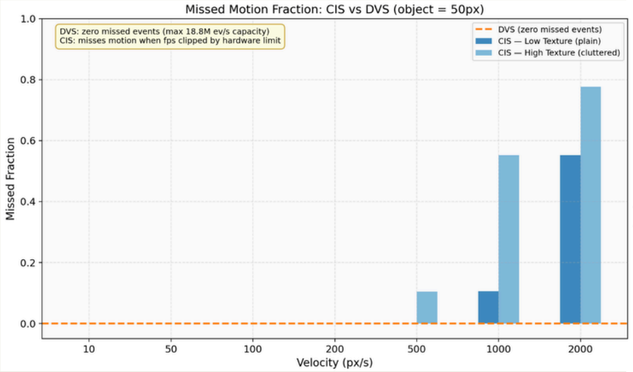
TRACKING MODEL AND ANALYSIS

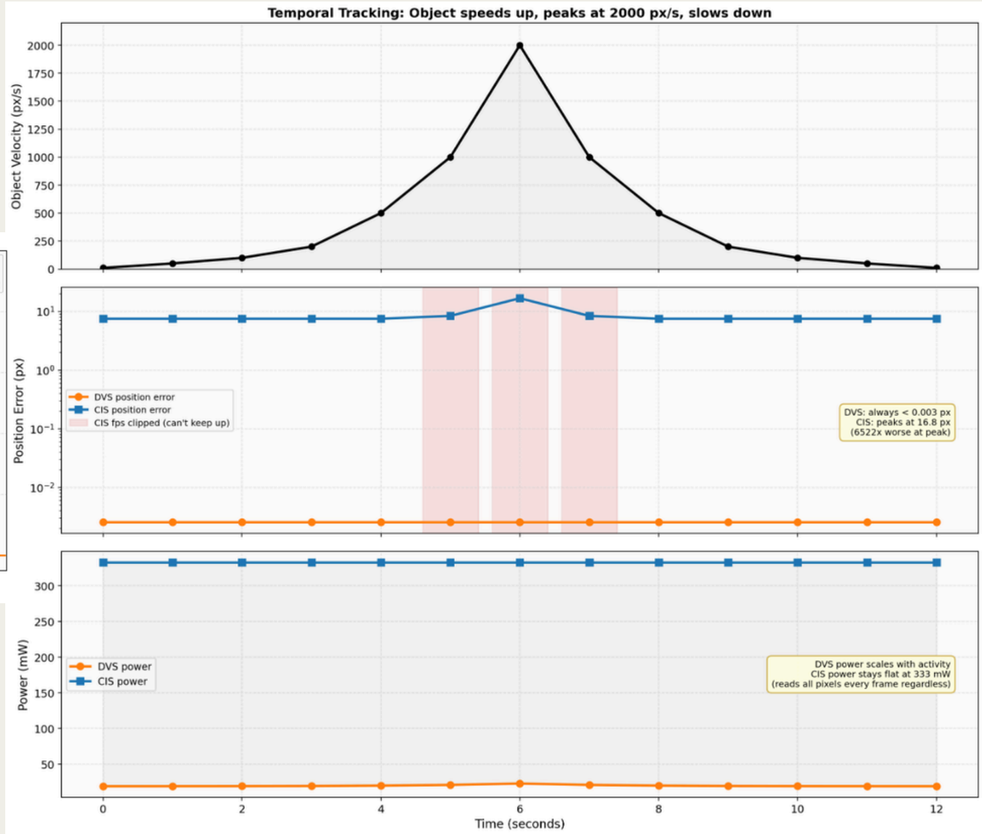
Background activity modeled as independent per-pixel Poisson events swept from 0% (static) to 50% (extreme) to bound DVS performance under real-world conditions where backgrounds are never perfectly still

Tracking Position Error: How far the object moves between consecutive sensor updates.

- **CIS position error =** inter-frame displacement + motion blur **= 1.5 × v / fps**
- **DVS position error =** inter-event displacement **= v / event_rate**







Thank you!

TRACKING ERROR EXAMPLE

Example: 50px ball moving at 500 px/s

CIS (frame-based):

- Hardware max frame rate: 179 fps
- Required fps for tracking: 500 / (50 × 0.1) = 100 fps → within limit, so actual fps = 100
- Frame time: 1/100 = 10 ms
- In those 10 ms, the ball moves: 500 × 0.01 = 5.0 pixels (inter-frame displacement)
- During exposure, the ball smears: 0.5 × 500 × 0.01 = 2.5 pixels (motion blur)
- Total position error: 5.0 + 2.5 = 7.5 pixels
- The tracker knows where the ball was at t=0 ms and t=10 ms, but has no idea what happened in between!

DVS (event-driven):

- Event rate at this scene: 194,480 events/sec
- Time between events: 1/194,480 = 0.005 ms (5 microseconds)
- In those 5 μs, the ball moves: 500 × 0.00000514 = 0.003 pixels
- No motion blur (no exposure period)
- Total position error: 0.003 pixels